



Lecturer: Flemming Christensen

INSTITUT FOR ELEKTRO- OG COMPUTERTEKNOLOGI, AARHUS UNIVERSITET

Email: flech@ece.au.dk

Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed.,  
<https://www.probabilitycourse.com>

## General Linear Regression

A regression model is called **linear** if it can be expressed in the form

$$y = \beta_0 f_0(x) + \beta_1 f_1(x) + \cdots + \beta_k f_k(x) + \varepsilon,$$

where  $f_0(x), \dots, f_k(x)$  are **known functions** of  $x$ , which may be **nonlinear** (e.g.,  $\sin(x)$ ,  $x^2$ ,  $\log(x)$ , ...).

The model is still considered *linear* as long as the **parameters**  $\beta_0, \beta_1, \dots, \beta_k$  **appear linearly** (i.e., they are not multiplied together, inside nonlinear functions, etc.).

$\varepsilon$  is an error term (noise term)

### ✓ Linear model examples:

- $y = \beta_0 + \beta_1 x$
- $y = \beta_0 + \beta_1 x^2$
- $y = \beta_0 + \beta_1 \log(x)$
- $y = \beta_0 + \beta_1 x + \beta_2 x^2$

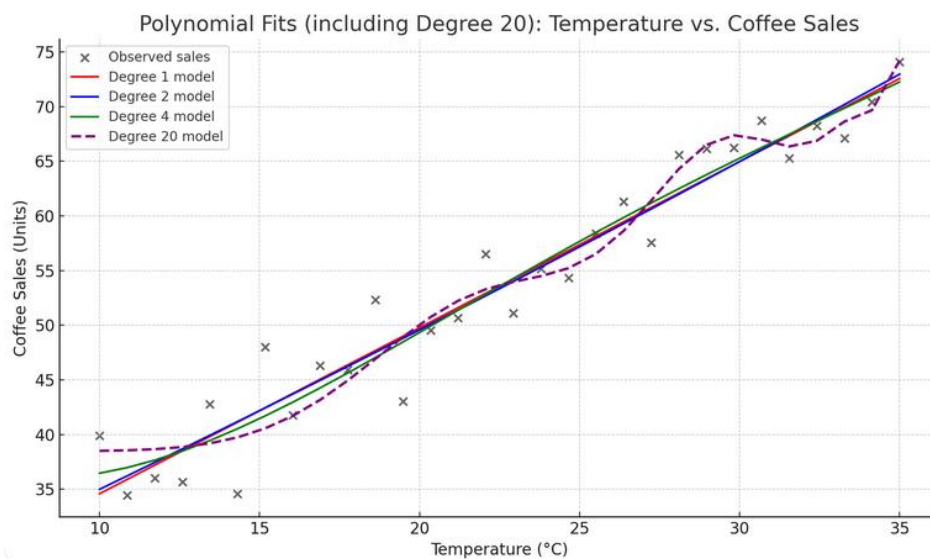
### ✗ Nonlinear model example:

- $y = \beta_0 + \beta_1^2 x$
- $y = \frac{\beta_0}{\beta_1 + x}$
- $y = \exp(\beta_0 + \beta_1 x)$

## General Linear Regression – overfitting?

**The simplest model is often the best**

**Example:** As the temperature increases, iced coffee sales also tend to increase.



The **degree 20 model fits the training data very tightly**, following even small fluctuations:

Those wiggles mean the model is:

1. **Fitting the noise**, not just the true underlying pattern.
2. **Reacting too much to outliers or random variation**, which is unlikely to generalize well to new (unseen) data.

**Conclusion:** Adding unnecessary complexity does not meaningfully improve predictions—and reduces interpretability.

## Simple Linear Regression

We consider the simplest case of a linear regression model.

**Simple linear regression:** Linear regression with a single predictor variable  $x$  and response variable  $y$

The statistical model is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

We observe  $n$  data points

$$(x_1, y_1), \dots, (x_n, y_n)$$

Assume that the relationship between  $x$  and  $y$  follows the linear regression model:

$$y_1 = \beta_0 + \beta_1 x_1 + \varepsilon_1$$

$$y_2 = \beta_0 + \beta_1 x_2 + \varepsilon_2$$

$$\vdots$$

$$y_n = \beta_0 + \beta_1 x_n + \varepsilon_n$$

Or in matrix notation

$$y = X\beta + \varepsilon$$

Diagram illustrating the matrix notation equation  $y = X\beta + \varepsilon$  with arrows pointing to the components:

- $y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$
- $X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}$
- $\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$
- $\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$

Solution formula

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

## Simple Linear Regression

### Interpretation

- $\beta_0$  : intercept
- $\beta_1$  : slope
- $\varepsilon_i$  : random **error term**

The errors represent influences on  $y$  that are **not explained by**  $x$ .

### Parameter Estimation

The parameters  $\beta_0, \beta_1$  are **unknown** and must be estimated from the data.

We compute estimates

$$\hat{\beta}_0, \hat{\beta}_1$$

### Errors

Deviation from the **true regression line**

$$\varepsilon_i = y_i - (\beta_0 + \beta_1 x_i).$$

Errors are **random variables** and cannot be observed.

We typically assume:

$$\varepsilon_1, \dots, \varepsilon_n \text{ are independent}$$

And

$$\varepsilon_i \sim N(0, \sigma^2)$$

## Simple Linear Regression

### The Fitted Regression Line

After estimating the parameters we obtain the **fitted regression line**

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

The **predicted value** for observation  $i$  is

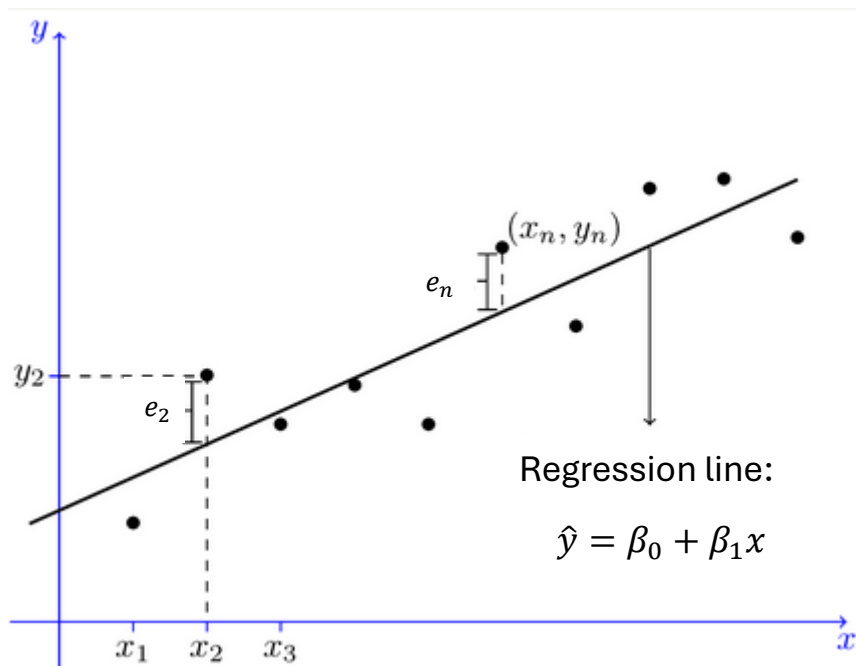
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

### Residual

Deviation from the **fitted regression line**

$$e_i = y_i - \hat{y}_i.$$

Residuals are observable and serve as **estimates of the errors**.



## Finding Model Parameters

### Goal

We estimate the parameters by choosing numbers

$$\hat{\beta}_0, \hat{\beta}_1$$

so that the fitted line

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

is as close as possible to the data.

### Sum of Squared Errors

Define

$$S(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

We choose  $\hat{\beta}_0, \hat{\beta}_1$  to **minimize** this function.

Take derivatives:

$$\frac{\partial S}{\partial \hat{\beta}_0} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\frac{\partial S}{\partial \hat{\beta}_1} = -2 \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

⇓

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

## Finding Model Parameters

### Expand

$$\begin{aligned}\sum y_i &= n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i \\ \sum x_i y_i &= \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2\end{aligned}$$

### Introduce sample means

$$\bar{x} = \frac{1}{n} \sum x_i, \quad \bar{y} = \frac{1}{n} \sum y_i$$

Divide the first equation by  $n$ :

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

so

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

### Substitute into the second equation

After simplification,

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

### Define convenient notation

$$s_{xx} = \sum (x_i - \bar{x})^2, \quad s_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$$

Then

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

## Finding Model Parameters

### Fitted regression line

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

### Residuals

$$e_i = y_i - \hat{y}_i$$

and

$$\sum_{i=1}^n e_i = 0$$

## Sum the residuals = 0

### Sum the residuals

$$\sum_{i=1}^n e_i = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i).$$

Expand:

$$= \sum y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum x_i.$$

### Use the formula for the intercept

From least squares we obtained

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Thus

$$\sum e_i = n\bar{y} - n(\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 n\bar{x}.$$

### Simplify

$$= n\bar{y} - n\bar{y} + n\hat{\beta}_1 \bar{x} - n\hat{\beta}_1 \bar{x}.$$

Everything cancels:

$$\sum_{i=1}^n e_i = 0.$$

Because of this,

- positive residuals above the line
- negative residuals below the line

## How good is the linear model?

- We can always find the best regression straight line to a set of datapoints  $(x_i, y_i)$
- But it doesn't mean that the regression straight line fits the datapoints very well – ie. that the linear model is a good model

| Question                                                    | Tool                |
|-------------------------------------------------------------|---------------------|
| Is there a <b>linear relationship</b> between $x$ and $y$ ? | t-test on $\beta_1$ |
| How much variation in $y$ is explained?                     | $R^2$               |
| Is the <b>linear form</b> appropriate?                      | Residual plots      |

## $R^2$

Assume the **population model**

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Taking variance gives

$$\text{Var}(Y) = \beta_1^2 \text{Var}(X) + \text{Var}(\varepsilon)$$

Interpretation:

- $\beta_1^2 \text{Var}(X)$  = variation in  $Y$  explained by  $X$
- $\text{Var}(\varepsilon)$  = unexplained variation (noise)

The fraction of variance explained by the regression is

$$R^2 = \frac{\text{explained variation}}{\text{total variation}} = \frac{\beta_1^2 \text{Var}(X)}{\text{Var}(Y)}$$

## $R^2$ for simple linear regression

Rewrite

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$$

where

$$s_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \quad s_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

and define

$$s_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2.$$

Then

$$R^2 = \frac{\left(\frac{s_{xy}}{s_{xx}}\right)^2 s_{xx}}{s_{yy}}.$$

Simplifying gives

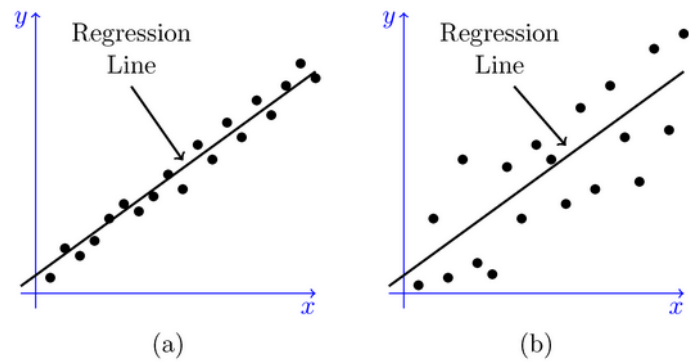
$$R^2 = \frac{s_{xy}^2}{s_{xx} s_{yy}}.$$

### Interpretation

- $0 \leq R^2 \leq 1$
- $R^2 \approx 1$ : regression line fits well (points close to line)
- $R^2 \approx 0$ : regression line explains little of the variation in  $Y$

**Quiz**

Which one has the highest  $R^2$  value?



**1: a**

**2: b**

**3:  $R^2$  not defined in either case**

## Residual plots

### Purpose

Residual plots are used to check whether the **assumptions of the linear regression model** are reasonable.

Assume the model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

where

$$E[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2.$$

After fitting the model we obtain the **fitted values**

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

and define the **residuals**

$$e_i = y_i - \hat{y}_i.$$

Residuals represent the **observed deviations from the fitted regression line**.

Ideally, residuals should behave like **random noise centered around zero**.

You assess two key properties:

#### 1. Linearity assumption

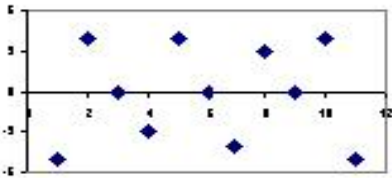
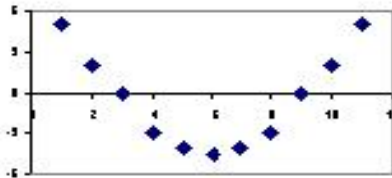
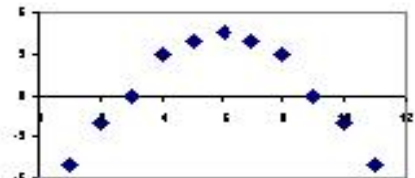
- Residuals should be randomly scattered around zero (no visible pattern).
- Patterns such as U-shaped or inverted U-shaped curves indicate that the true relationship is **nonlinear**, meaning a simple linear model is inappropriate.

#### 2. Homoscedasticity (constant variance)

- The spread (variance) of residuals should be roughly the same for all  $x_i$ .
- A "funnel" or increasing/decreasing spread suggests **heteroscedasticity**, which violates model assumptions and affects inference reliability.

**Quiz**

Which one has the most appropriate linear form?

**a****b****c**

**1: a**

**2: b**

**3: c**

## Statistical inference on linear regression

### Hypothesis test

We test whether there is a **linear relationship** between  $X$  and  $Y$ .

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

If

$$\beta_1 = 0$$

then the expected value of  $Y$  does **not depend on  $X$** , meaning there is **no linear relationship**.

## Statistical inference on linear regression

Point estimator:  $\hat{\beta}_1$

From least squares, the estimated slope is

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}},$$

which serves as a **point estimator** of the true parameter  $\beta_1$ .

Like any estimator based on sample data,  $\hat{\beta}_1$  is a **random variable** — it varies from one sample to another.

### Under the standard regression assumptions

- Errors  $\varepsilon_i$  are independent and normally distributed with mean 0 and variance  $\sigma^2$ :

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

then the slope estimator  $\hat{\beta}_1$  itself follows a **normal distribution**:

$$\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{s_{xx}}\right).$$

## Statistical inference on linear regression

### Estimating the variance

Since  $\sigma^2$  (the population error variance) is unknown, we estimate it from the data using the **residuals**:

$$s_r^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$


n-2 degrees of freedom

Each residual  $e_i = y_i - \hat{y}_i$  depends on the fitted regression line.

But that line itself was **estimated** from the data by fitting **two parameters**:

- the intercept  $\beta_0$
- the slope  $\beta_1$

Because those two parameters were estimated from the same  $n$  data points, we “use up” 2 degrees of freedom.

### Constructing the test statistic

We now use this variance estimate to standardize the slope estimate:

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{s_r^2 / s_{xx}}}.$$

Under the null hypothesis ( $H_0 : \beta_1 = \beta_{1,0}$ ), this statistic follows a **Student's t-distribution** with  $n - 2$  degrees of freedom.

## Statistical inference on linear regression

### Decision rule and p-value

The **p-value** is the probability, under  $H_0$ , of observing a  $t$ -value as extreme (or more) than the one you computed:

$$p\text{-value} = 2 (1 - t_{\text{cdf}}(|t|, n - 2)) .$$

- If  $p < \alpha$  (e.g. 0.05), reject  $H_0$ .
- This means there is evidence of a linear relationship between  $X$  and  $Y$ .

## Example: Hubble's law

- Hubble's law is the name for the observation in physical cosmology that objects observed in deep space are found to have a relative velocity away from the Earth that is approximately proportional to their distance from the Earth:  
 $v = H \cdot x$
- Edwin Hubble's original measurements for 24 distant galaxies were:

```
Distance = [ 0.032 0.034 0.214 0.263 0.275 0.275 ...
             0.450 0.500 0.500 0.630 0.800 0.900 ...
             0.900 0.900 0.900 1.000 1.100 1.100 ...
             1.400 1.700 2.000 2.000 2.000 2.000 ];
```

```
Speed = [ 170 290 -130 -70 -185 -220 200 290 ...
          270 200 300 -30 650 150 500 920 ...
          450 500 500 960 500 850 800 1090 ];
```

## The Model

$$v = \beta_0 + \beta_1 x + \varepsilon$$

where:

- $v$ : speed (km/s) — dependent variable
- $x$ : distance (Mpc) — independent variable
- $\varepsilon$ : random error term
- $\beta_0$ : intercept (velocity when distance = 0)
- $\beta_1$ : slope (the rate of change of speed with distance; the **Hubble constant**)

## Example: Hubble's law

### Estimating the Parameters

We estimate  $\beta_0, \beta_1$  from sample data  $(x_i, y_i)$  using the **least-squares method**.

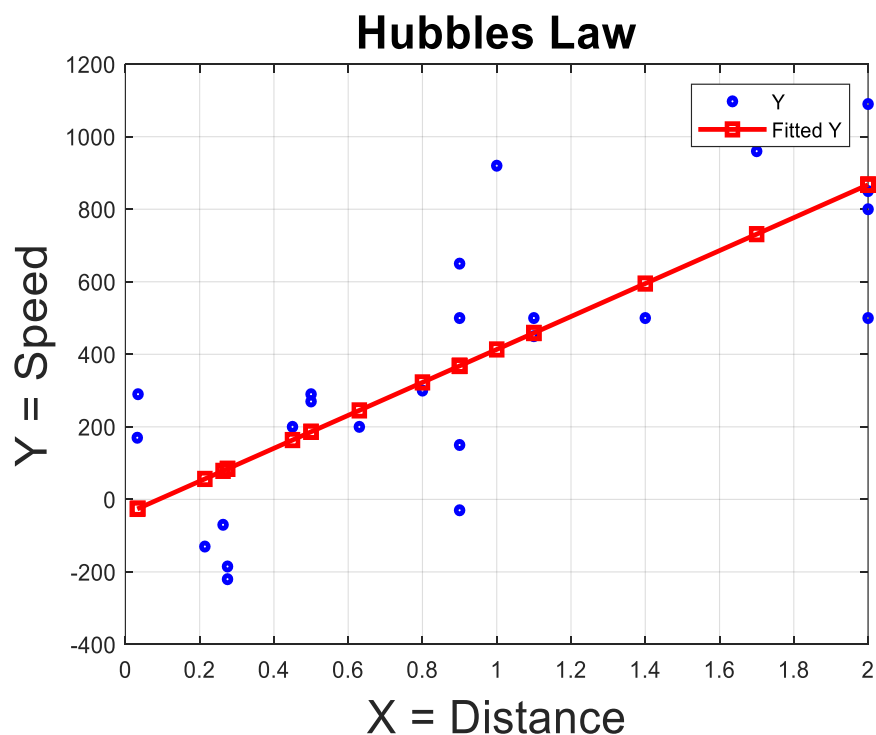
$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

For Hubble's original data:

$$\hat{\beta}_1 = 454.16, \quad \hat{\beta}_0 = -40.78$$

That is, the estimated line is:

$$\hat{y} = -40.78 + 454.16x$$



$R^2$  is typically between 0.9 and 0.99.

## Example: Hubble's law

### Statistical Inference on the Slope ( $\beta_1$ )

We now test whether the relationship between distance and velocity is *statistically significant*.

Hypotheses:

$$H_0 : \beta_1 = 0 \quad (\text{no relationship})$$

$$H_1 : \beta_1 \neq 0 \quad (\text{significant linear relationship})$$

Sampling distribution:

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{s_r^2}{s_{xx}}\right)$$

where

$$s_r^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad s_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

Test statistic:

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{s_r^2 / s_{xx}}} \sim t(n-2)$$

*p*-value:

$$p\text{-value} = 2(1 - t_{cdf}(|t|, n-2))$$

where  $T_{n-2}$  follows a Student's *t*-distribution with  $n - 2$  degrees of freedom.

## Example: Hubble's law

### Statistical Inference on the Slope ( $\beta_1$ ) ..

Result for Hubble's data:

$$t = 6.035, \quad p \approx 0$$

Since  $p < 0.05$ , we reject  $H_0$  and conclude:

There is a statistically significant linear relationship between galaxy distance and velocity — supporting Hubble's Law.

### Confidence Interval for the Slope

The 95% confidence interval for  $\beta_1$  is:

$$\beta_1 \in \left[ \hat{\beta}_1 - t_0 \sqrt{\frac{s_r^2}{s_{xx}}}, \hat{\beta}_1 + t_0 \sqrt{\frac{s_r^2}{s_{xx}}} \right]$$

where

$$t_0 = t_{cdf}^{-1}(0.975, n - 2)$$

is the **critical value** from the *Student's t-distribution* with  $n - 2$  degrees of freedom — the value for which the central 95% of the distribution lies between  $-t_0$  and  $+t_0$ .

For the Hubble data:

$$t_0 = 2.0739, \quad \beta_1 \in [298.12, 610.19]$$

Since **0 is not within the interval**, this confirms that the slope is **significantly different from 0** — supporting the presence of a linear relationship between distance and velocity.

## Example: Hubble's law

### Final Interpretation (Hubble's Law Example)

- The **slope** ( $\beta_1$ ) is **significantly positive** → distant galaxies move away faster.
- The **intercept** ( $\beta_0$ ) is **not significantly different from zero** → when distance  $\approx 0$ , speed  $\approx 0$ .
- Both results support Hubble's Law:

$$v = H_0 \times \text{distance}$$

where  $H_0 = \hat{\beta}_1 \approx 454.16 \text{ km/s/Mpc}$ .

## Simple Linear Regression Model

### Example

We are given the data points:

$$(1, 3), \quad (2, 4), \quad (3, 8), \quad (4, 9).$$

We want to:

1. Find the estimated regression line  $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$  and plot it.
2. Compute the coefficient of determination  $R^2$ .
3. Plot the residuals and comment on their behavior.
4. Test whether  $X$  and  $Y$  are linearly related at the 5% significance level using the  $p$ -value for  $\hat{\beta}_1$ .

**Calculation of parameters by hand:**

## Method 1: Using standard formulas

We use

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

and

$$s_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \quad s_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

**Compute the sample means**

$$\bar{x} = \frac{1 + 2 + 3 + 4}{4} = \frac{10}{4} = 2.5$$

$$\bar{y} = \frac{3 + 4 + 8 + 9}{4} = \frac{24}{4} = 6$$

**Compute  $s_{xx}$** 

$$s_{xx} = \sum_{i=1}^4 (x_i - \bar{x})^2$$

Compute each term:

$$(1 - 2.5)^2 = (-1.5)^2 = 2.25$$

$$(2 - 2.5)^2 = (-0.5)^2 = 0.25$$

$$(3 - 2.5)^2 = (0.5)^2 = 0.25$$

$$(4 - 2.5)^2 = (1.5)^2 = 2.25$$

Add them:

$$s_{xx} = 2.25 + 0.25 + 0.25 + 2.25 = 5$$

**Compute  $s_{xy}$**

$$s_{xy} = \sum_{i=1}^4 (x_i - \bar{x})(y_i - \bar{y})$$

Compute each term:

$$(1 - 2.5)(3 - 6) = (-1.5)(-3) = 4.5$$

$$(2 - 2.5)(4 - 6) = (-0.5)(-2) = 1$$

$$(3 - 2.5)(8 - 6) = (0.5)(2) = 1$$

$$(4 - 2.5)(9 - 6) = (1.5)(3) = 4.5$$

Add them:

$$s_{xy} = 4.5 + 1 + 1 + 4.5 = 11$$

**Compute the slope**

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} = \frac{11}{5} = 2.2$$

**Compute the intercept**

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Substitute the values:

$$\hat{\beta}_0 = 6 - (2.2)(2.5)$$

$$\hat{\beta}_0 = 6 - 5.5 = 0.5$$

**Result from Method 1**

$$\hat{\beta}_0 = 0.5, \quad \hat{\beta}_1 = 2.2$$

So the estimated regression line is

$$\hat{y} = 0.5 + 2.2x$$

**Calculation of parameters by hand:**

Method 2: Using linear algebra formula

We now use

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

where

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}, \quad y = \begin{pmatrix} 3 \\ 4 \\ 8 \\ 9 \end{pmatrix}$$

and

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}.$$

**Compute  $X^T$**

$$X^T = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

**Compute  $X^T X$**

$$X^T X = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}$$

$$X^T X = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}$$

Compute  $X^\top y$

$$X^\top y = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 8 \\ 9 \end{pmatrix}$$

$$X^\top y = \begin{pmatrix} 24 \\ 71 \end{pmatrix}$$

Compute  $(X^\top X)^{-1}$

We need the inverse of

$$\begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}.$$

For a  $2 \times 2$  matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

provided  $ad - bc \neq 0$ .

Here

$$a = 4, \quad b = 10, \quad c = 10, \quad d = 30$$

so the determinant is

$$ad - bc = (4)(30) - (10)(10) = 120 - 100 = 20$$

Thus

$$(X^\top X)^{-1} = \frac{1}{20} \begin{pmatrix} 30 & -10 \\ -10 & 4 \end{pmatrix}$$

Compute  $\hat{\beta} = (X^\top X)^{-1} X^\top y$

$$\hat{\beta} = \frac{1}{20} \begin{pmatrix} 30 & -10 \\ -10 & 4 \end{pmatrix} \begin{pmatrix} 24 \\ 71 \end{pmatrix}$$

$$\hat{\beta} = \frac{1}{20} \begin{pmatrix} 10 \\ 44 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 2.2 \end{pmatrix}$$

**Result from Method 2**

$$\hat{\beta} = \begin{pmatrix} 0.5 \\ 2.2 \end{pmatrix}$$

So again

$$\hat{\beta}_0 = 0.5, \quad \hat{\beta}_1 = 2.2$$

and the regression line is

$$\hat{y} = 0.5 + 2.2x$$

## Complete Solution By Python:

```
python

import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# Data
x = np.array([1, 2, 3, 4])
y = np.array([3, 4, 8, 9])

# --- 1. Regression coefficients ---
n = len(x)
x_bar, y_bar = np.mean(x), np.mean(y)
s_xx = np.sum((x - x_bar)**2)
s_xy = np.sum((x - x_bar)*(y - y_bar))

beta1_hat = s_xy / s_xx
beta0_hat = y_bar - beta1_hat * x_bar

# Fitted values and residuals
y_hat = beta0_hat + beta1_hat * x
residuals = y - y_hat

# --- 2. R^2 calculation ---
s_yy = np.sum((y - y_bar)**2)
R2 = s_xy**2 / (s_xx * s_yy)

# --- 3. Test for significance of slope (H0:  $\theta_1 = 0$ ) ---
s_r2 = np.sum(residuals**2) / (n - 2)
se_beta1 = np.sqrt(s_r2 / s_xx)
t_stat = beta1_hat / se_beta1
p_value = 2 * (1 - stats.t.cdf(abs(t_stat), df=n-2))
```

## Solution By Python:

```
# --- 4. Print results ---
print(f" $\beta_0$  = {beta0_hat:.3f}")
print(f" $\beta_1$  = {beta1_hat:.3f}")
print(f" $R^2$  = {R2:.3f}")
print(f" $t$  = {t_stat:.3f}, p-value = {p_value:.4f}")

# --- 5. Plot regression line ---
plt.figure(figsize=(6,4))
plt.scatter(x, y, color="blue", label="Data")
plt.plot(x, y_hat, color="red", label="Fitted line")
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Simple Linear Regression")
plt.legend()
plt.grid(True)
plt.show()

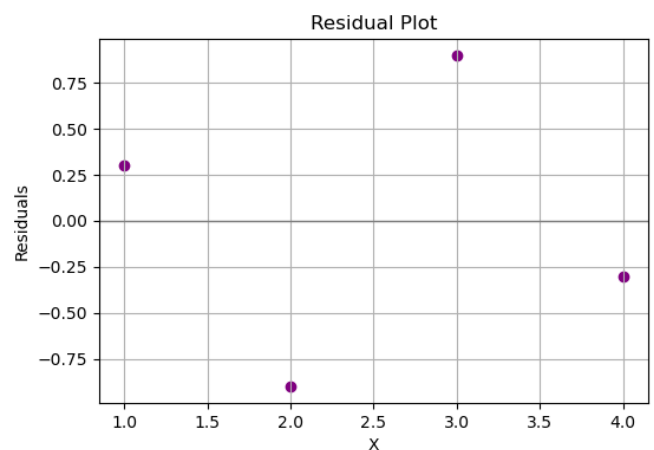
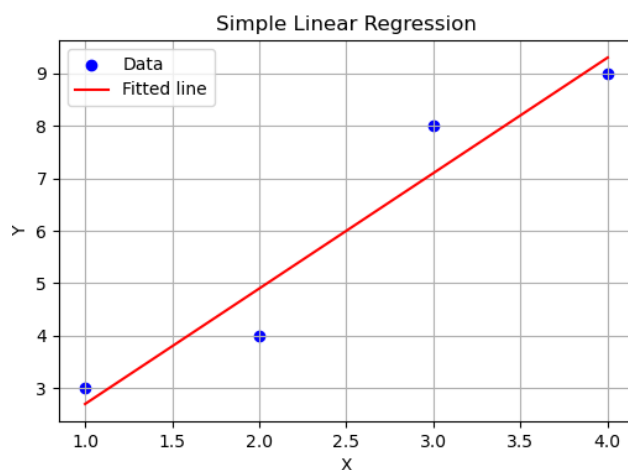
# --- 6. Residual plot ---
plt.figure(figsize=(6,4))
plt.axhline(0, color='gray', linewidth=1)
plt.scatter(x, residuals, color='purple')
plt.xlabel("X")
plt.ylabel("Residuals")
plt.title("Residual Plot")
plt.grid(True)
plt.show()
```

## Solution By Python:

| Quantity                     | Symbol          | Value |
|------------------------------|-----------------|-------|
| Intercept                    | $\hat{\beta}_0$ | 0.5   |
| Slope                        | $\hat{\beta}_1$ | 2.2   |
| Coefficient of determination | $R^2$           | 0.931 |
| t-statistic (df = 2)         | $t$             | 5.19  |
| p-value                      | $p$             | 0.035 |

So the estimated regression line is

$$\hat{y} = 0.5 + 2.2x$$



## Interpretation

- $R^2 = 0.931$ 
  - About **93.1% of the variation in  $Y$**  is explained by  $X$ .
- $p = 0.035 < 0.05$ 
  - The slope is **statistically significant at the 5% level**.
  - We **reject  $H_0 : \beta_1 = 0$** .
- The residuals appear roughly centered around zero, suggesting the **linear model is reasonable** for this small dataset.